Department of Computer Science and Engineering

Academic Year: 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3551 & Distributed Computing

Faculty member (s): Mrs. P. Devisri, AP/CSE

Innovative Practice Description Unit

Topic: Unit V / Suzuki Kasami Algorithm

Course Outcome: CO 4

Topic Learning Outcome: 4b

Activity Chosen: Peer Teaching Activity

Justification:

The Suzuki–Kasami Algorithm is an important concept in Distributed Systems, used to achieve mutual exclusion in distributed environments. As the topic involves understanding token passing, request queues, and critical section management, peer teaching helps students break down the complex workflow into simpler, understandable steps. By allowing students to take the role of teachers, the activity encouraged collaborative learning, improved conceptual clarity, and enabled students to analyze how distributed coordination works in real-world systems. Peer interaction provided various perspectives and strengthened the overall learning process, making the algorithm easier to visualize and apply.

Time Allotted for the Activity: 35 minutes

Details of the Implementation:

- Before the session, the teacher asked for volunteers willing to teach concepts related to the **Suzuki–Kasami Token-Based Distributed Mutual Exclusion Algorithm**. A few students showed interest.
- Based on their interest, **Harikrishnan** and **Kasmi** were assigned to teach the topic.
- Students received the topic one day earlier to prepare explanations, examples, and simple demonstrations.
- During the session:

  Harikrishnan explained the basic idea of the algorithm, including request numbers, the token, and the concept of a unique shared token in a distributed system and demonstrated a **real-time token-passing role-play** using students as nodes. They used a paper "Token" and request number cards to show how the token moves between nodes when entering the critical section.
- Students enacted the algorithm:

  Each student acted as a node (Node 1, Node 2, Node 3…).

  Request numbers were shown using flashcards.

  The token was passed to the node with the highest priority request (as per the queue).
- Students clearly understood the role of the request vector, queue mechanism, and how critical sections are accessed without conflict.
- Question-and-answer time was given at the end of the session. Students asked doubts related to token loss, queue updates, concurrency issues, and real-time use cases of the algorithm.

CO – PO / PSO mapping:

CO	PO1	PO2	PO10	PSO1
CO 5	2	2	1	1

(1 – Low 2 – Moderate 3 High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO10**	**PSO1**
	2	2	1	1
Justification for correlation	Students understood the core concepts of distributed mutual exclusion and the Suzuki–Kasami algorithm.	Students understood the core concepts of distributed mutual exclusion and the Suzuki–Kasami algorithm.	Students communicated effectively by presenting complex algorithmic processes such as critical section entry, token generation, and queue updates.	Students gained the ability to apply distributed algorithms to real-time software systems and cloud-based coordination problems.

Images / Screenshot of the practice:

Hari Krishnan

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

- Students showed great interest during the demonstration and actively participated in the role-play.
- They felt that the algorithm, which usually seems difficult in theory, became much easier to understand through the live token-passing activity.

• Students expressed increased confidence in explaining distributed algorithms, queues, and token-based synchronization.
• The activity improved their communication skills and helped them simplify complex algorithmic concepts during peer discussions..

❖ ***Benefit of the practice:***

• Students gained deeper conceptual clarity of the Suzuki–Kasami algorithm through experiential learning.
• Active involvement improved memory retention compared to conventional teaching methods.
• The activity enhanced communication and analytical skills.
• Students engaged in self-learning and explored real-time use cases such as distributed databases, cloud coordination, and resource access control.
• The activity motivated students to share knowledge with classmates confidently.

❖ ***Challenges faced in implementation:***

- Motivating shy students to participate in the role-play activity required extra effort.
- Preparing the token cards, request numbers, and role assignments took more time than regular teaching.
- Ensuring that all students clearly understood queue updates during the activity required repeated explanations.

References:

1. https://effectiviology.com/protege-effect-learn-by-teaching
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Stud ent_S oft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-023/4 Collaborative_Coding_DBMS_KVS.pdf